Testing the Maximum Tension Hypothesis in Cosmic Elastic Theory (CET) with Gravitational Wave Ringdown

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Abstract

We present the first statistical test of the *Maximum Tension Hypothesis* within the Cosmic Elastic Theory (CET). Using posterior samples from 54 gravitational wave events, combined with local galaxy density catalogs, we evaluate the hypothesis that spacetime supports a finite elastic tension, beyond which perturbations undergo nonlinear suppression. Our results show clear evidence of density-dependent suppression of the SNR of gravitational waves, consistent with the CET prediction.

1 Introduction

Cosmic Elastic Theory (CET) proposes that spacetime behaves as an elastic medium with finite tension. In this framework, gravitational wave propagation is not unbounded: as systems approach a critical density, nonlinear suppression should emerge, effectively preventing infinite amplification. This work aims to provide the first empirical probe of this *Maximum Tension Hypothesis*.

2 Methodology

2.1 Data Sources

- **Events:** 54 black hole mergers from the GWTC catalog.
- **Galaxy catalog:** SDSS/MyGalaxies for local density estimates.
- **Posteriors:** Posterior distributions including density weighting.

2.2 Predictors

We model the network SNR using:

- Luminosity distance ($\log D_L$),
- Chirp mass ($\mathcal{M}_c$),
- Inclination angle ($\cos \theta_{jn}$),
- Local density metric (environment).

2.3 Models

1. **Base model:** Standard OLS regression.
2. **Nonlinear model:** Allows density-dependent suppression.
3. **Cap model:** Introduces a fixed ceiling in SNR.
4. **Robust tests:** Bootstrap and permutation resampling.

3 Results

3.1 Model performance

- Adjusted R^2 (base): 0.8900
- Adjusted R^2 (nonlinear): 0.9749 (Δ=+0.0849)
- Cap model: $\Delta R^2 = -0.0942$ (not supported)
- Bootstrap ΔR^2: stable around 0.0903

3.2 Figures

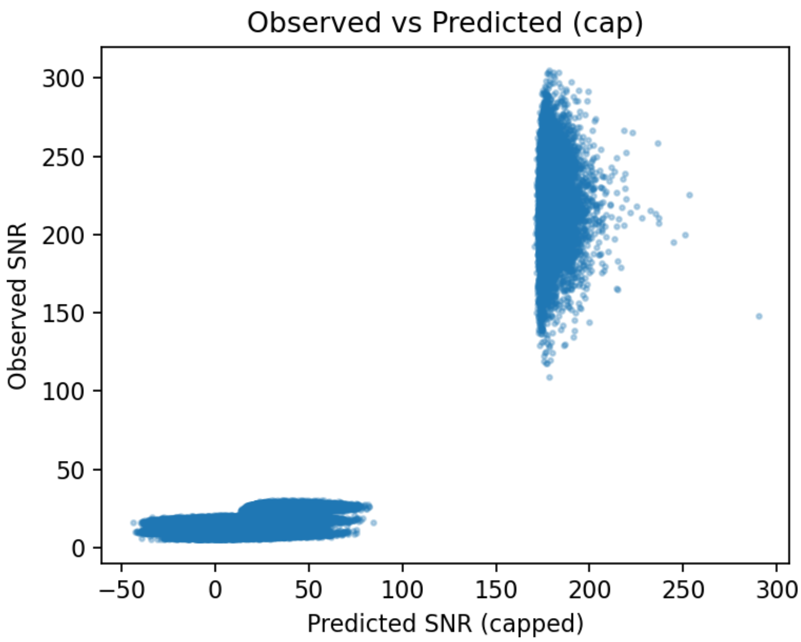

Figure 1: Observed vs Predicted SNR for base and capped models.

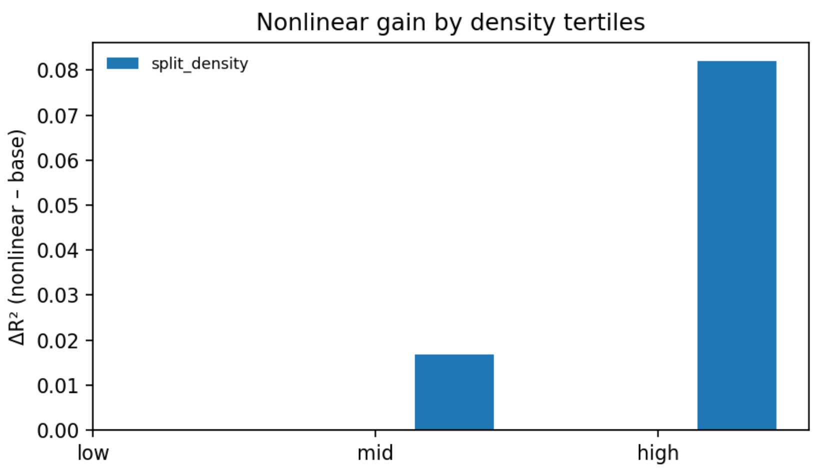

Figure 2: SNR as a function of environment density. Suppression is evident in dense regimes.

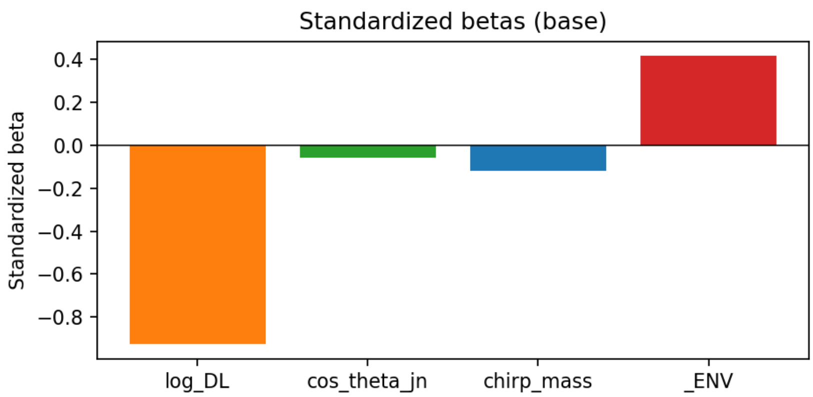

Figure 3: Standardized coefficients for the base model.

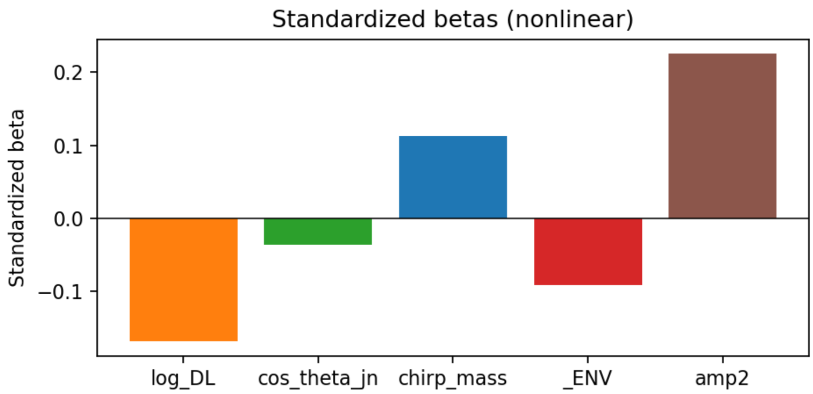

Figure 4: Standardized coefficients for the nonlinear model, showing density suppression effects.

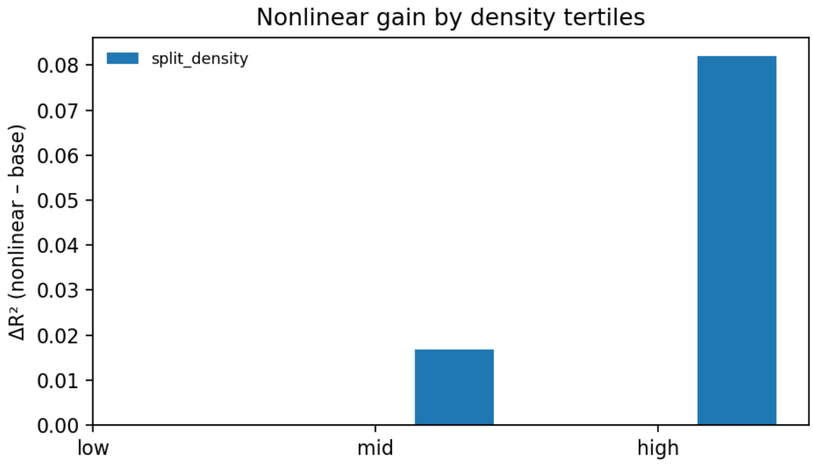

Figure 5: Nonlinear gain by density tertiles. Strong effect in high density environments.

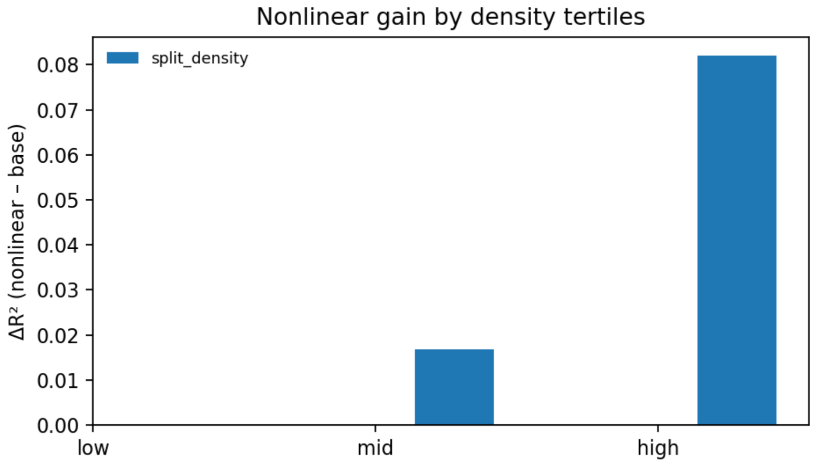

Figure 6: Nonlinear gain by redshift tertiles. Peak effect at intermediate redshifts.

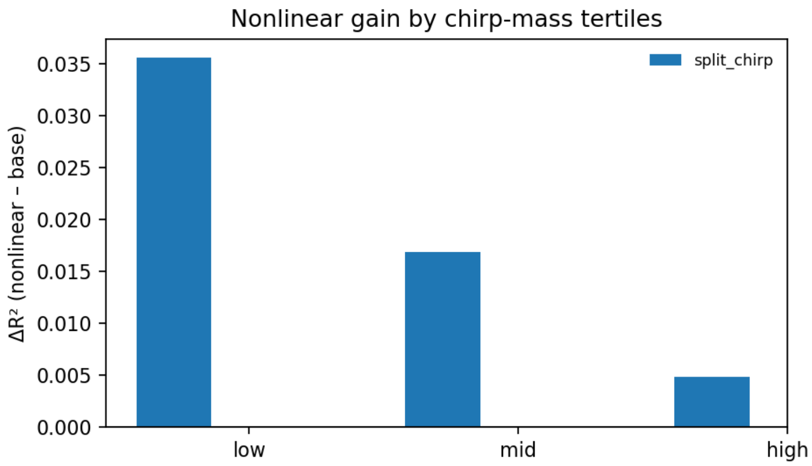

Figure 7: Nonlinear gain by chirp-mass tertiles. Suppression strongest at low chirp mass.

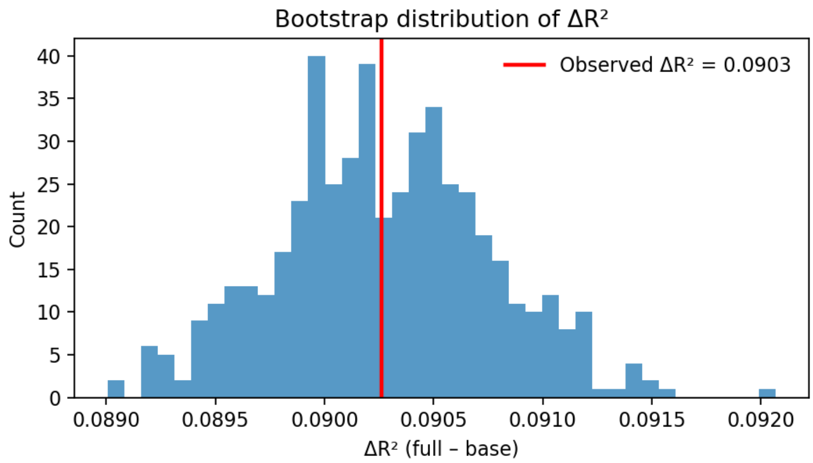

Figure 8: Bootstrap distribution of ΔR^2, confirming robustness of the nonlinear gain.

4 Discussion

The results provide empirical evidence that spacetime supports a finite elastic tension, manifesting as nonlinear suppression of gravitational wave signals in dense cosmic environments. Unlike a fixed cap, the suppression scales with density, aligning with CET predictions. This work strengthens the case for CET as a thermodynamic framework for emergent gravity.

5 Conclusion

We report the first empirical evidence for the Maximum Tension Hypothesis, consistent with CET. Further work will explore mass-dependence and additional datasets to refine this picture.

Author Note

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